9 A company has to make a large number of boxes.

The company has 6 machines.
All the machines work at the same rate.
When all the machines are working, they can make all the boxes in 9 days.

The table gives the number of machines working each day.

	day 1	day 2	day 3	all other days
Number of machines working	3	4	5	6

Work out the total number of days taken to make all the boxes.

Total work = 6 × 9 = 54 machine-days.

1 mark
Use 54 machine-days.

First 3 days = 3 + 4 + 5 = 12.

Work left = 54 − 12 = 42 machine-days.

1 mark
Remaining 42 at 6/day.

At 6 machines: 42 ÷ 6 = 7 more days.

Total = 3 + 7 = 10 days.

1 mark
Total 10 days.

10

(Total for Question 9 is 3 marks)